

Education

Carnegie Mellon University , Pittsburgh, PA	2024
PhD candidate in Computer Science. Advised by Professors Deva Ramanan and Mahadev Satyanarayanan. GPA: 4.2/4.0	(expected)
Stanford University , Stanford, CA	06/18
MS in Computer Science. GPA: 3.7/4.0	
Stanford University , Stanford, CA	06/12
BS in Computer Science , minor in Management Science and Engineering. GPA: 3.4/4.0	

Professional Experience

Engineering Manager and Technical Lead , Palantir Technologies, New York, NY	08/12-08/19
- Led R&D efforts to improve Palantir's Foundry data platform: - Created interactive search and analysis infrastructure that evolved into one of the primary user interfaces of Palantir's Foundry data lake platform deployed across the company's customer fleet - Invented full-text search system that powers Palantir's internal log analysis infrastructure and a significant part of the Foundry data lake platform - Led inception and rollout of one of Palantir's flagship data analytics products - Managed the technical implementation of a variety of high-profile customer engagements in the commercial sector (leading teams of 10-100+ individuals) - Supervised and mentored a team of 7+ direct reports	
Cofounder , TradeSensei, Palo Alto, CA	01/12-08/12
- Founded a trading startup and wrote the backend infrastructure enabling beta users to participate in real-time trading competitions - Participated in Stanford StartX accelerator program	
Software Engineering Intern , Addepar, Mountain View, CA	04/11-09/11
Worked on developing a software platform to help private financial advisors and family offices perform quantitative analysis, better visualize financial data, understand risk, and interface with their clients	

Research Experience

Graduate Research Assistant, Dept. of Computer Science , Carnegie Mellon University	08/19-Present
Conducting edge-computing and computer vision related research under the guidance of Professor Mahadev Satyanarayanan	
Graduate Independent Study , Stanford University, Stanford, CA	01/18-06/18
- Conducted research under Professor Alex Aiken and contributed to the Legion programming system - Implemented the Barnes-Hut simulation in a novel parallel programming language and benchmarked it against the Sherlock supercomputing cluster	
Research Assistant , Stanford University, Stanford, CA	06/10-12/10
- Conducted computer vision research under Professor Daphne Koller - Co-authored two accepted conference papers	

Teaching Experience

- Graduate Student Instructor, Dept. of Computer Science, Carnegie Mellon University** 09/20-
Present
- Teaching assistant for 15-640 (Distributed Systems) and inaugural 17-700 class (Data Science and Machine Learning at Scale)
 - Support and mentor team of undergraduate teaching assistants
 - Create and grade assignments, exams, and class projects
 - Design and present lecture materials
 - Support students during office hours and class forums
- Project Mentor, Dept. of Computer Science, Carnegie Mellon University** 09/20-
Present
- Designed a term project for 15-821 (Mobile and Pervasive Computing)
 - Meet and provide guidance to project members on a regular basis

Publications (Reverse Chronological Order)

- [1] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan, "Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs." Under submission.
- [2] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan, "OpenRTiST: End-to-End Benchmarking for Edge Computing," in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, 1 Oct.-Dec. 2020
- [3] M. Satyanarayanan, T. Eiszler, J. Harkes, **H. Turki** and Z. Feng, "Edge Computing for Legacy Applications," in IEEE Pervasive Computing, vol. 19, no. 4, pp. 19-28, 1 Oct.-Dec. 2020
- [4] M. P. Kumar, **H. Turki**, D. Preston and D. Koller, "Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, 1 July 2015.
- [5] M. P. Kumar, **H. Turki**, D. Preston and D. Koller, "Learning specific-class segmentation from diverse data," 2011 International Conference on Computer Vision, Barcelona, 2011, pp. 1800-1807.

Patents

- [1] **Haithem Turki**, Robert Fink, Amr Al Mallah. "Systems and Methods for Indexing and Searching." US Patent 2019/0347343 A1, filed June 8, 2018, and issued November 14, 2019.
- [2] **Haithem Turki**, Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara. "Systems and Methods for Constraint Driven Database Searching." US Patent 2018/293239 A1, filed April 11, 2017, and issued October 11, 2018.

Additional Information

- **Volunteer Activities:** Alchemist Accelerator (mentor and investor)
- **Languages:** Fluent in French and Arabic. Currently learning Japanese and German.
- **Other Qualifications:** Eberly Future Faculty Program. Top Secret security clearance (current). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.